Advantages of constant mesh gear box

1. Gear changing is simple
2. Less noise
3. Less chance for the teeth of the gears to break up.
4. Smoothness in operation
5. Low maintenance

Disadvantages

1. For changing gears from one speed to other speed, double de clutching has to be done.
2. More wear
3. Possibility of wear in dog clutch.

3.5.3 Synchromesh gear box

Synchromesh gear box is similar in construction as constant mesh gear box but differs slightly. Instead of dog clutch used in constant mesh gear box, a synchronizing unit is used in synchromesh gear box. In the synchromesh mesh gear box the gears 6, 7 and 8 in the output shaft are always in mesh with the gears 5, 4 and 3 in the lay shaft. In the clutch shaft and in the lay shaft there are gears 1 and 2 respectively which are always in mesh with each other. The gears in the main shaft are supported by bushes. Gear 5 in the lay shaft is in mesh with an idler gear. After the clutch gear in the clutch shaft and before the reverse gear in the main shaft there are synchronizers (S_2 and S_1 respectively) present. These synchronizers are allowed to move on the slots made on the main output shaft. All the shafts in the gear box are supported by the bearings in the housing.

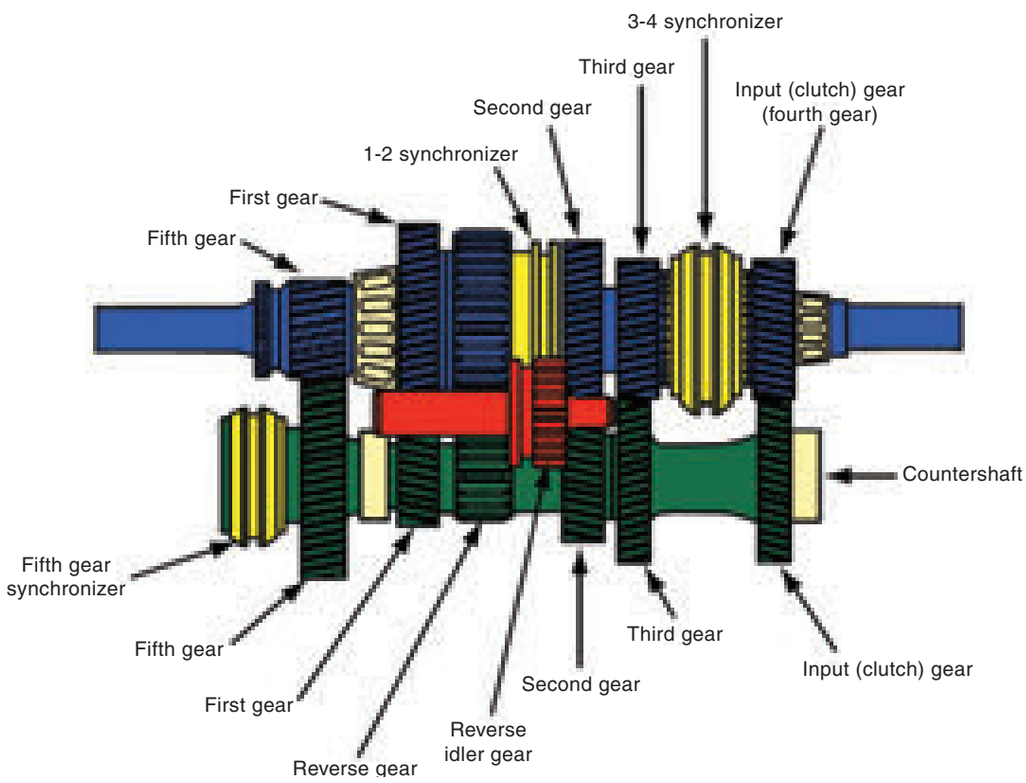


Figure 3.5.3 (a) A view of gear arrangements of a 5 speed synchromesh gear box

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